Boston Bus Equity

October Project Check-In Report, CS506 / Spark!
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Key terms

Term	What it means
Timepoint	A bus stop where the MBTA checks whether a bus is keeping to its schedule. A route has only a few of these.
Crossing	One bus passing one timepoint. A single trip creates several crossings.
On time	A crossing that passes the MBTA's reliability test. The on-time rate is the share of crossings that pass.
Target routes	15 priority routes from the Livable Streets report: 14, 15, 17, 22, 23, 24, 26, 28, 29, 31, 33, 42, 44, 45, 111.

Data Processing

What are our data sources, and how did we collect them?

All of our data is public, and our notebook downloads it automatically.

Data	Where it came from	How we got it	What we used it for
Bus arrival and departure times, 2020, 2024	MBTA	Downloaded files	Reliability, wait, trip time
Bus ridership by route, Fall 2016-2025	MBTA	Downloaded files	Number of riders on each route
Rider survey, 2025 (latest public)	MBTA	API	Riders' race and income
Rider survey, 2015-17	Planning agency (CTPS)	Downloaded file	Comparison with earlier riders
Bus stop locations	MBTA	Downloaded file	Placing each stop on the map
Census income and race, 2020-24	Census Reporter	API	Who lives near each stop
Census tract boundaries	US Census	API	Matching stops to tracts
Service Delivery Policy, Fall 2024	MBTA	Read the policy	The definition of on time

What cleaning and processing did we do, and why?

What we did	Why
Compared Jan-Feb 2020 with Jan-Feb 2024	These are the last pre-COVID months available, and using the same months removes seasonal effects.
Kept weekdays only, and stopped 2024 at November	Weekend schedules differ, and the network changed in December 2024.
Matched route names across files (SL1 = 741)	Files name routes differently, so busy routes would otherwise be lost.
Left out 1.7 million blank arrival times	A blank can be a canceled trip or a data gap. Counting them all as not on time lowers the 2024 on-time rate from 66.3% to 61.4%.
Removed 2,025 extreme delays (over 30 min early or 2 hours late)	These are likely tracking errors.
Matched bus stops to census tracts	This links bus performance to neighborhood income and race.

Check: our Fall 2018 ridership is within 14% of the Livable Streets figures on all 15 target routes.

Why did we choose these features?

Feature	Why
On-time rate	This is the MBTA's own reliability standard.
Average delay	It shows how late buses run, which the on-time test can miss on frequent routes.
Expected wait	It estimates how long riders wait at the stop.
Trip time	It shows how long trips take and how predictable they are.
Weekday boardings	They show how many people each route affects.
Race and income	They show who is affected, both for riders (survey) and for neighborhoods (census).

Data Visualizations

Figure 1. Ridership over time

Verdict: Supported: ridership has not fully recovered, reaching 79% of its Fall 2019 level in Fall 2024 and 80% in Fall 2025.

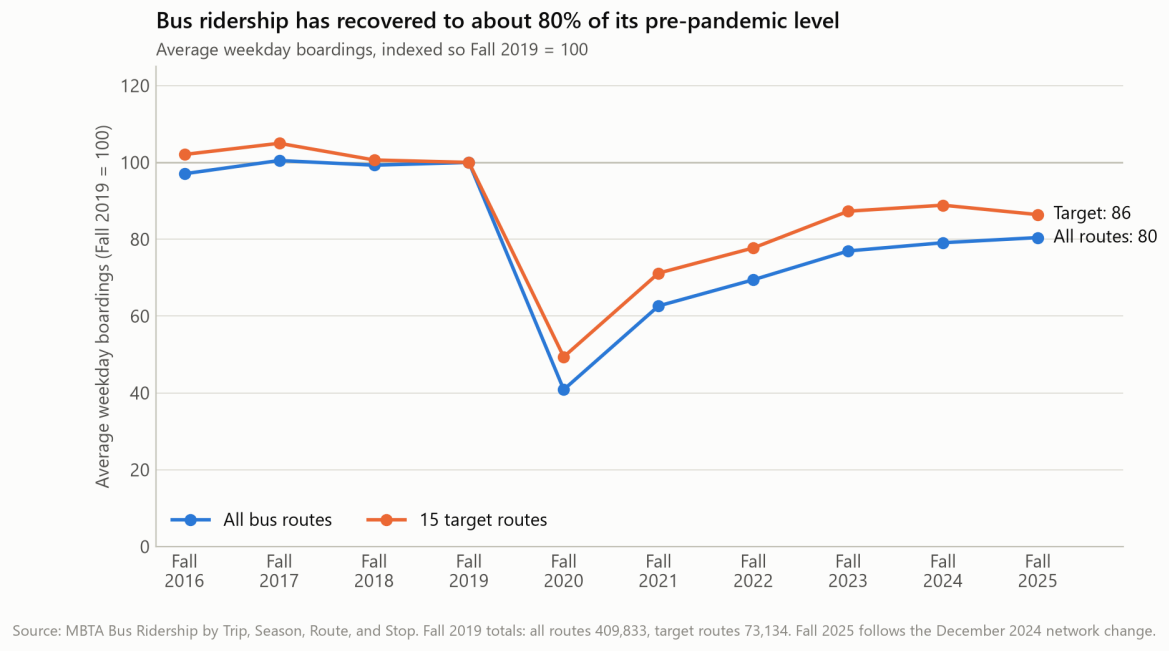


Figure 2. Ridership by route

Verdict: Supported: recovery is uneven, with a median route change of -20% and only 18 of 127 routes gaining riders.

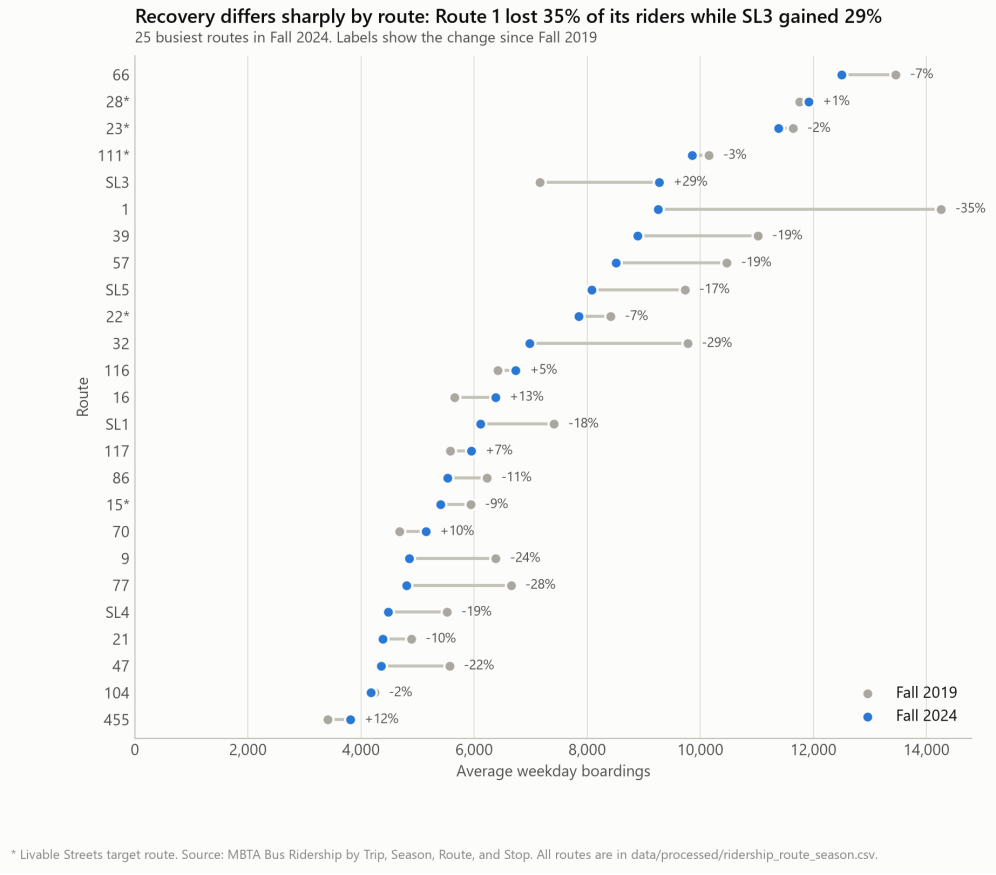
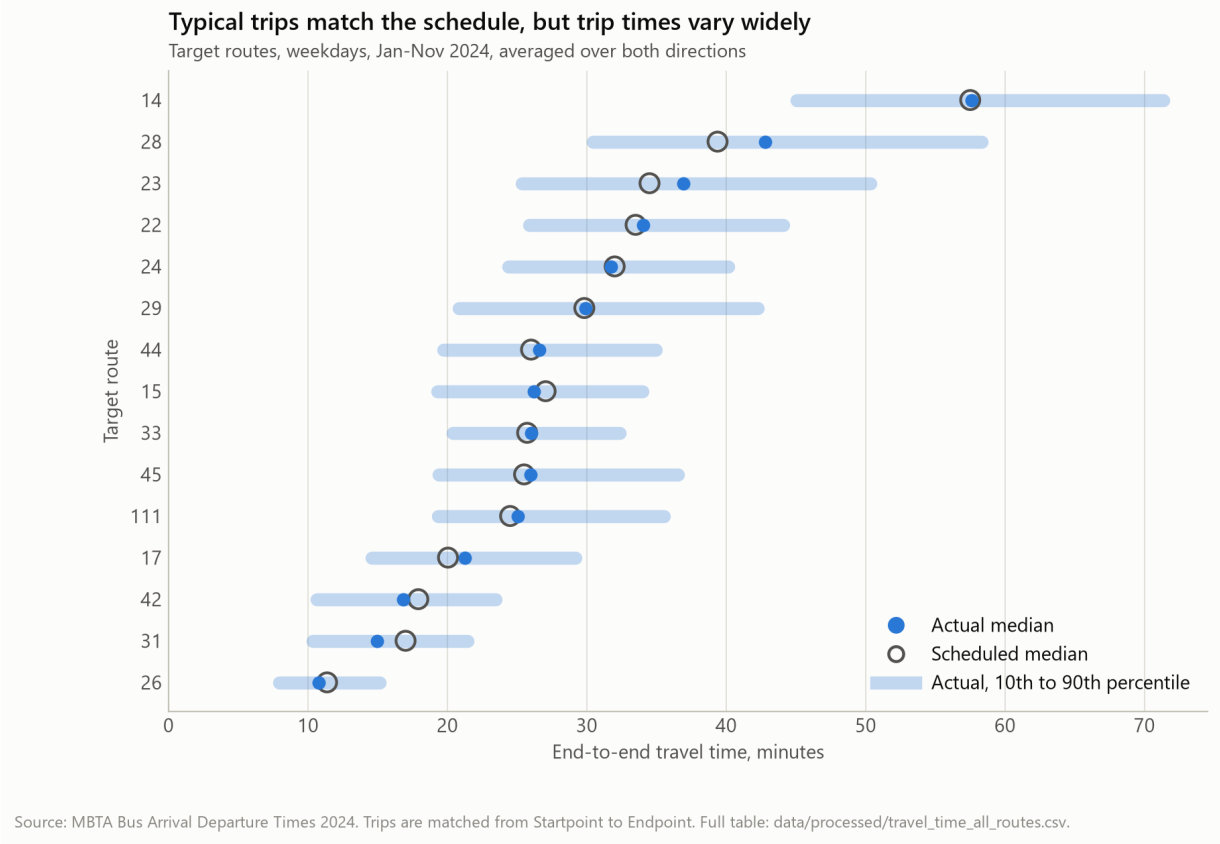


Figure 3. End-to-end travel time

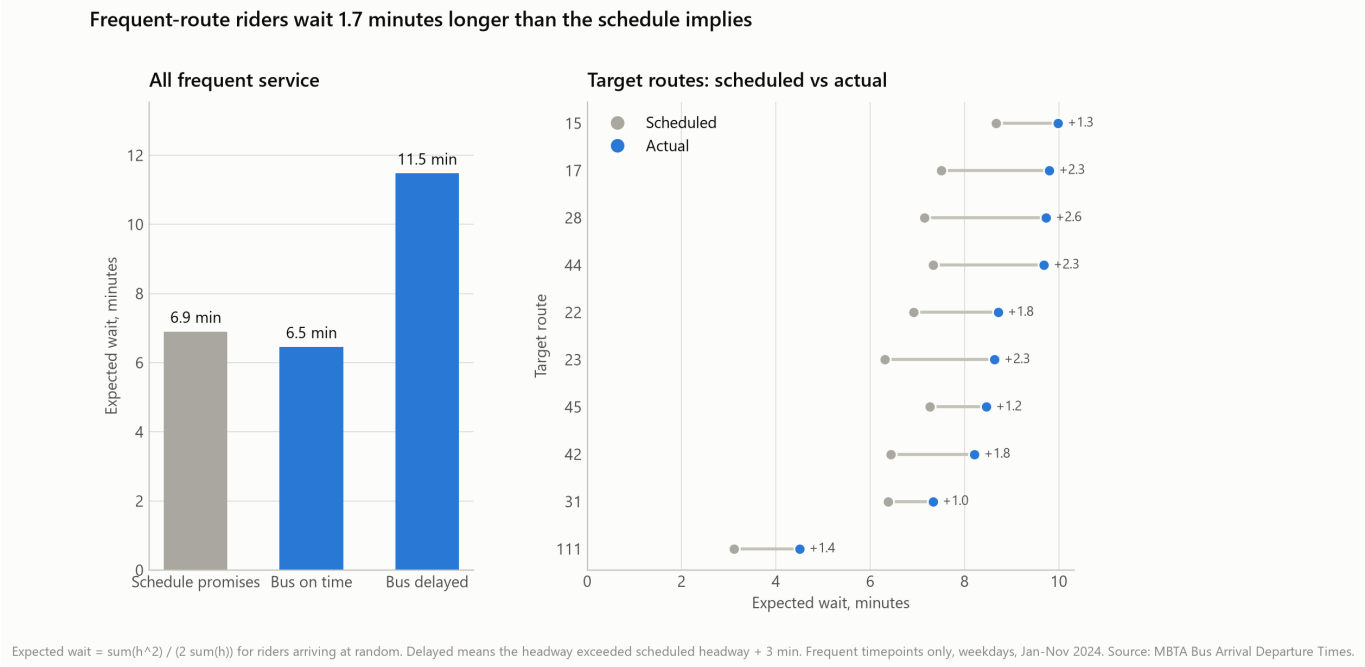
Verdict: Partly supported: the median route is only 0.1 minutes over schedule, but its fast and slow trips differ by 16.4 minutes.



Source: MBTA Bus Arrival Departure Times 2024. Trips are matched from Startpoint to Endpoint. Full table: data/processed/travel_time_all_routes.csv.

Figure 4. Rider wait time

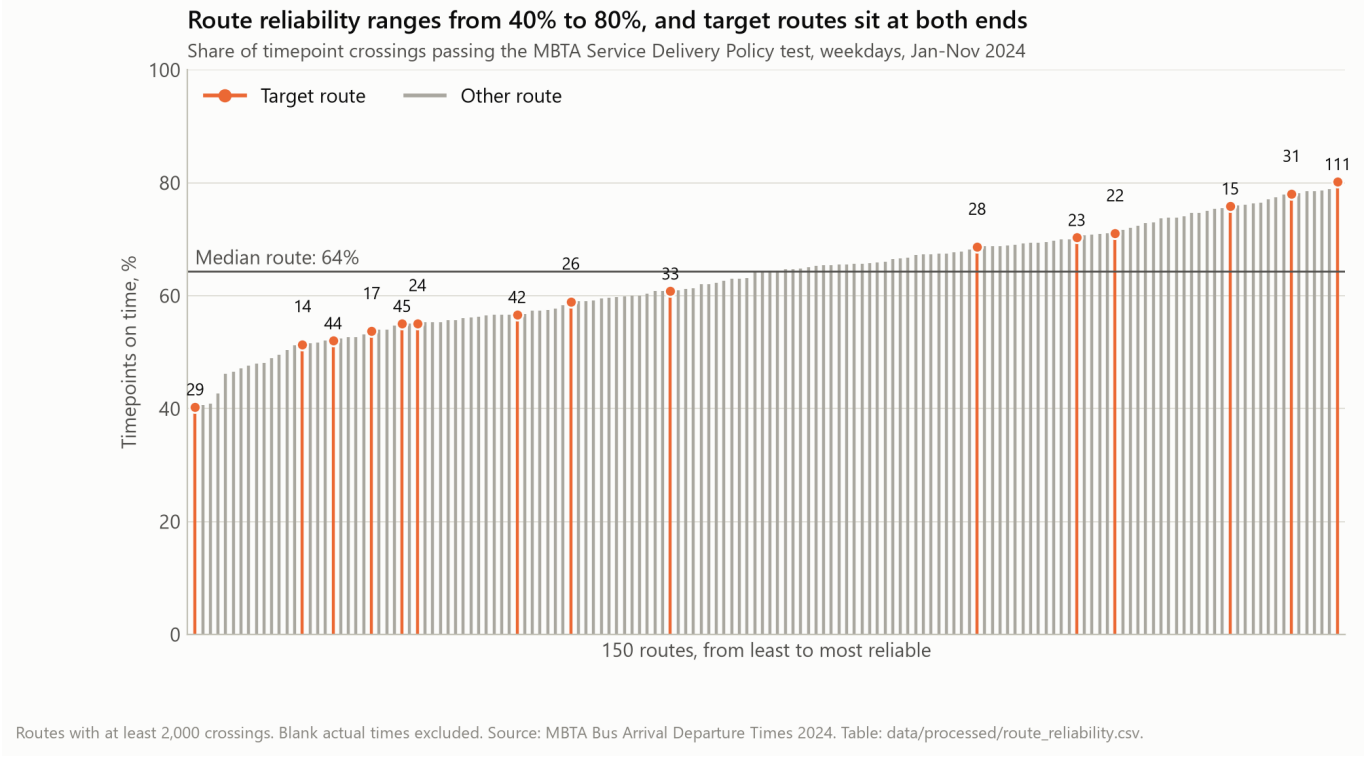
Verdict: Supported: frequent-route riders wait 8.6 minutes against 6.9 implied by the schedule, rising to 11.5 minutes when a bus is delayed.



Expected wait = sum(h^2) / (2 sum(h)) for riders arriving at random. Delayed means the headway exceeded scheduled headway + 3 min. Frequent timepoints only, weekdays, Jan-Nov 2024. Source: MBTA Bus Arrival Departure Times.

Figure 5. Reliability by route

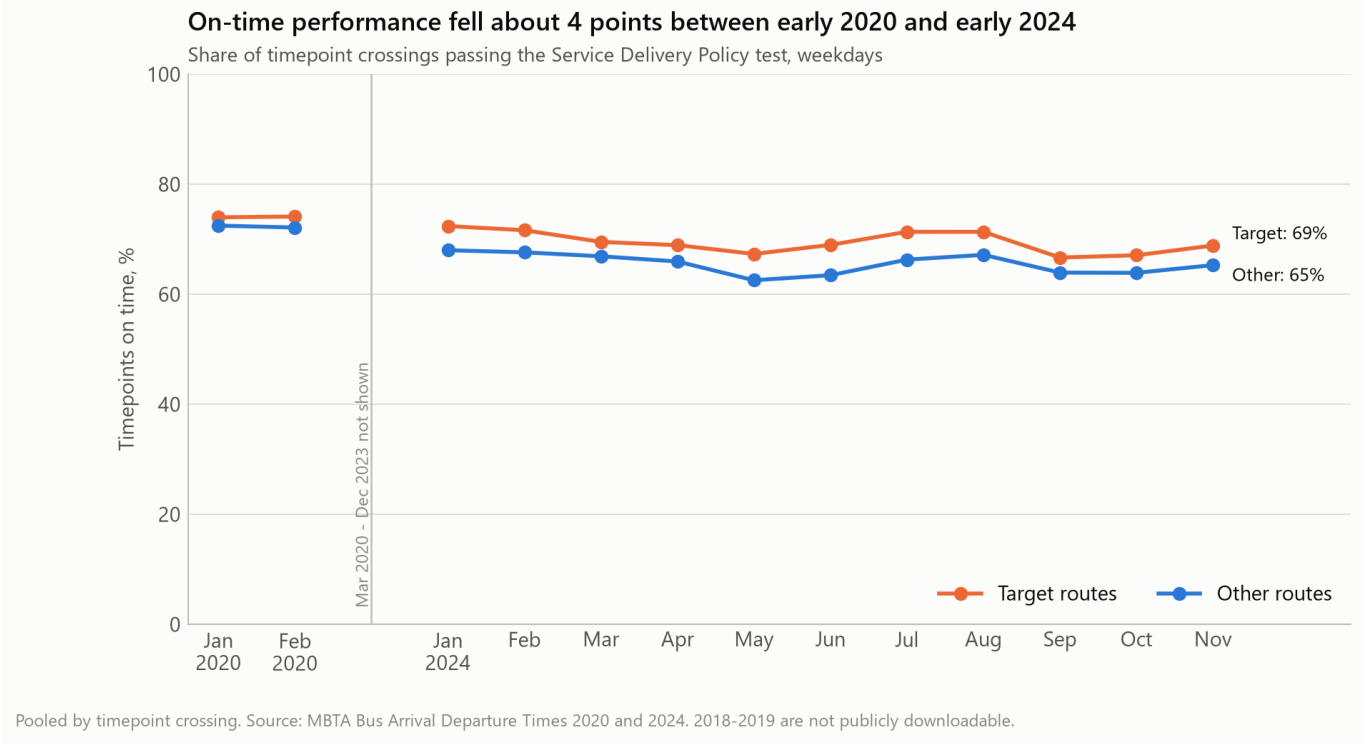
Verdict: Mixed: routes differ widely, from 40% to 80% on time, but target routes match other routes on this test (p = 0.59) while running 1.4 minutes later (p = 0.002).



Routes with at least 2,000 crossings. Blank actual times excluded. Source: MBTA Bus Arrival Departure Times 2024. Table: data/processed/route_reliability.csv.

Figure 6. Reliability over time

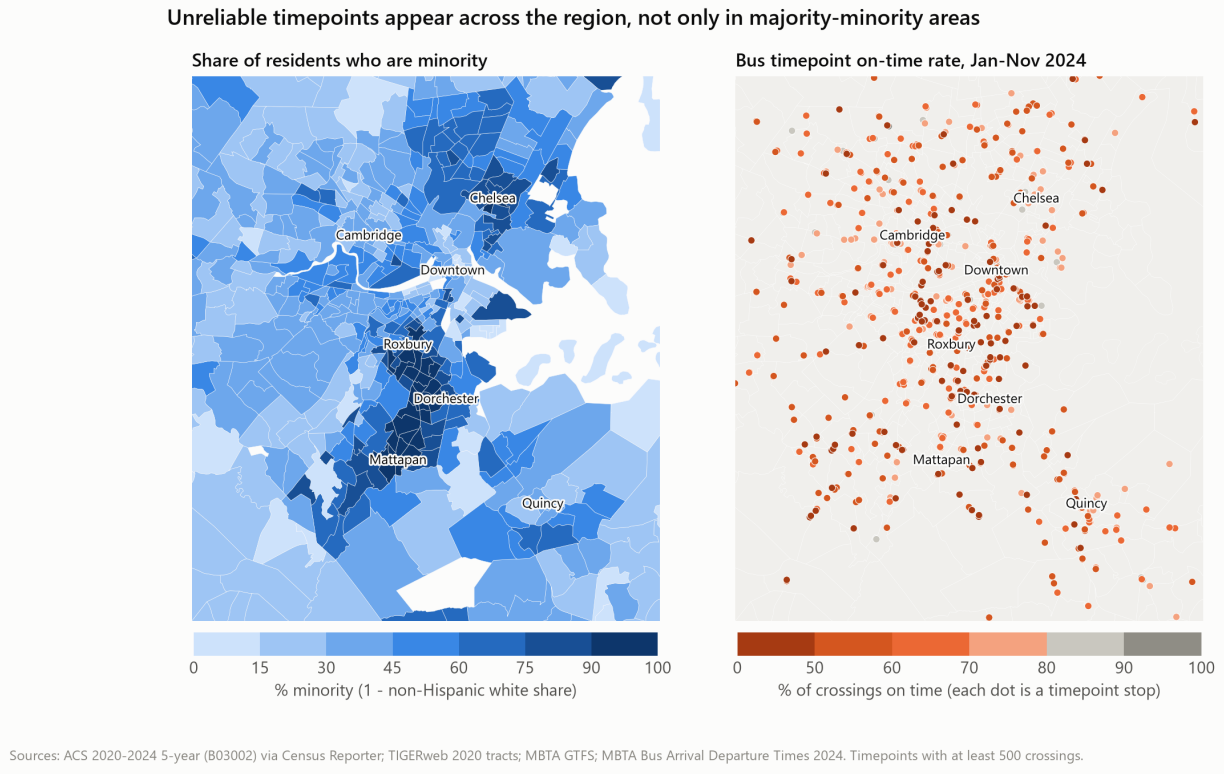
Verdict: Supported: from January-February 2020 to the same months of 2024, the on-time rate fell from 72.6% to 68.6% and mean delay rose from 2.26 to 3.85 minutes.



Pooled by timepoint crossing. Source: MBTA Bus Arrival Departure Times 2020 and 2024. 2018-2019 are not publicly downloadable.

Figure 7. Reliability and neighborhoods

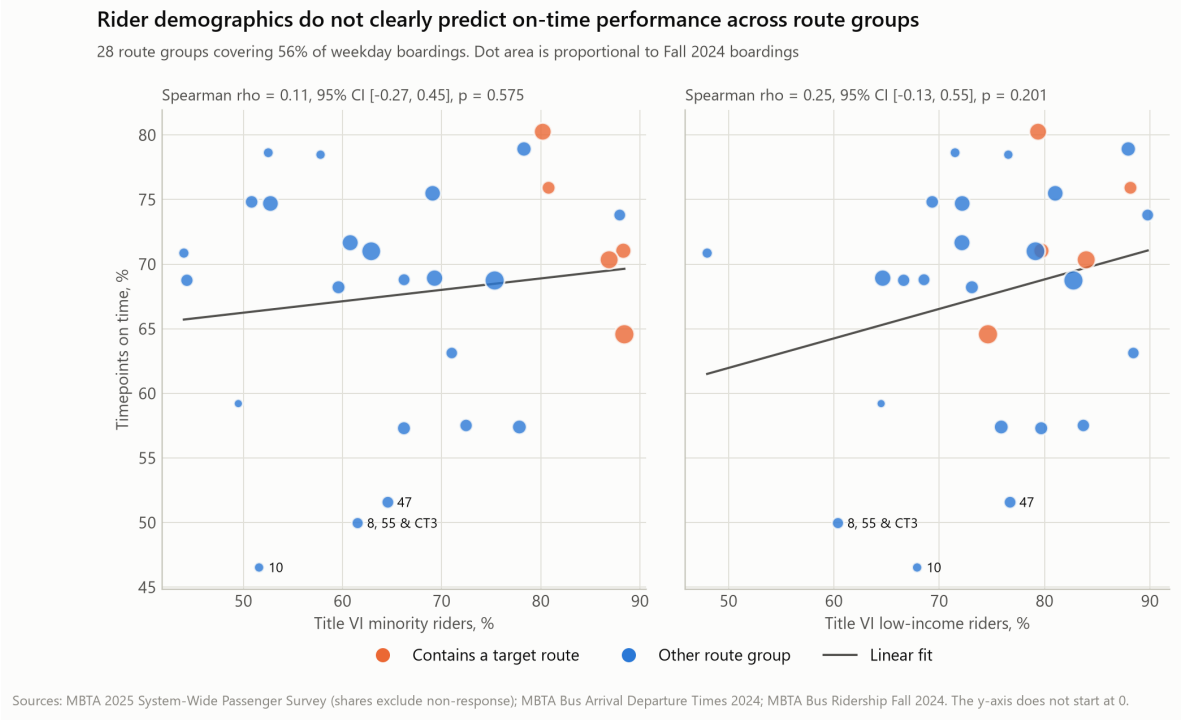
Verdict: Challenges a simple reading: 407 of 1,031 mapped timepoints are under 60% on time, and they appear throughout the region, not only in majority-minority areas.



Sources: ACS 2020-2024 5-year (B03002) via Census Reporter; TIGERweb 2020 tracts; MBTA GTFS; MBTA Bus Arrival Departure Times 2024. Timepoints with at least 500 crossings.

Figure 8. Rider demographics by route

Verdict: Not supported, but a weak test: minority share (rho = 0.11) and low-income share (rho = 0.25) have confidence intervals that include zero, across 28 route groups covering 56% of boardings.



Sources: MBTA 2025 System-Wide Passenger Survey (shares exclude non-response); MBTA Bus Arrival Departure Times 2024; MBTA Bus Ridership Fall 2024. The y-axis does not start at 0.

Figure 9. Neighborhood demographics by tract

Verdict: Partly supported: tract race and income show no link to the on-time test but a modest link to delay (rho = 0.25 with minority share, -0.19 with income).



Figure 10. Rider-hours lost

Verdict: Weakly supported: lost time is not highly concentrated, since the top 15 of 147 routes account for 45% of the estimated hours and 38% of boardings.

